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EXAM AWS CERTIFIED DEVELOPER - ASSOCIATE DVA-C02 TOPIC 1 QUESTIO...

Exam question from Amazon's AWS Certified Developer - Associate DVA-C02
Question #: 456
Topic #: 1
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An application ingests data from an Amazon Kinesis data stream. The shards in the data stream are set for normal traffic.

During tests for peak traffic, the application ingests data slowly. A developer needs to adjust the data stream to handle the peak traffic.

What should the developer do to meet this requirement MOST cost-effectively?

A. Install the Kinesis Producer Library (KPL) to ingest data into the data stream.

B. Switch to on-demand capacity mode for the data stream. Specify a partition key when writing data to the data stream.

C. Decrease the amount of time that data is kept in the data stream by using the DecreaseStreamRetentionPeriod API operation.

D. Increase the shard count in the data stream by using the UpdateShardCount API operation.

Show Suggested Answer

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by albert_kuo at *Nov. 18, 2024, 1:56 p.m.*

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Comments

YUICH 1 year, 3 months ago

Selected Answer: D

Comparison of B and D:
B (On-Demand Mode) is better when:

Traffic patterns are highly variable or unpredictable.
Automatic scaling is preferred to minimize manual intervention.
The workload is intermittent, making On-Demand pricing more cost-efficient.
D (Manually Adjusting Shard Count) is better when:

The application has predictable traffic peaks and troughs.
The goal is to manually control scaling and costs in provisioned mode.
Immediate scaling is required to meet throughput needs, as On-Demand Mode may introduce a slight delay for shard scaling.
Conclusion:
The correct solution depends on the specific requirements mentioned in the question. If the question highlights unpredictable traffic, On-Demand Mode (B) is likely the better choice. However, if the traffic peaks are predictable and manual control is desired, increasing shards (D) is the optimal solution.

upvoted 4 times

ShakthiGCP 1 year, 3 months ago

Selected Answer: B

On-demand - data streams with an on-demand mode require no capacity planning and automatically scale to handle gigabytes of write and read throughput per minute. With the on-demand mode, Kinesis Data Streams automatically manages the shards in order to provide the necessary throughput.

upvoted 1 times

ShakthiGCP 1 year, 3 months ago

The key word is 'Codt effective' - you dont want to pay more by updating the shards for higher rate.

upvoted 1 times

albert_kuo 1 year, 4 months ago

Selected Answer: D

aws kinesis update-shard-count \
--stream-name my-data-stream \
--target-shard-count 4 \
--scaling-type UNIFORM_SCALING

upvoted 2 times

YUICH 1 year, 4 months ago

Switching to on-demand capacity mode provides automatic scaling, which eliminates the need for manual adjustments while remaining cost-efficient.
Kinesis adjusts the shard count dynamically based on actual usage,

ensuring sufficient capacity during peak traffic without overprovisioning.

upvoted 1 times